§6 Minimal structure of ACS relations — part 4 of 4 (representative categories and six families, scope, hand-off to later sections, summary)

Source: `paper/section6_relational.tex`, lines 1792–2171 (`subsec:rel_categories` and `subsec:rel_phase1a`: the set of 18 representative category labels, the six-family classification, the undefined status of category conditions, derived and composite categories, categories and recognized relations, hand-off to §7, items left to Phase 1.α and Phase 2, hand-off to §7 and §8, and the summary of the descriptive policy of §6), based on commit e6e12cd, with the author's correction to the count statement near line 1824 recorded in ledger entry L-A3.195. Status: draft. Part 3 is `06_relational_3.md`. This part defines no assumption; the three assumptions of §6 are in parts 1 and 2. The Japanese manuscript is authoritative; this is a summary, not a translation, and adds no definition, assumption or result.

Representative categories and the six families

The subsection organizes relation categories along two axes: the level at which a category is established (unidirectional or bidirectional) and a classification by content.

Definition (`def:category_set`): set of representative category labels (classification C). The range of the category component of a unidirectional relation descriptor is

$\mathcal{C}_{\rightarrow}$:= {obs, ref, aff, dep, sup, par, dom, mgmt, manip, exploit, use, comp, alert, avoid, excl},

and the range of the categories established between ACSs bidirectionally is $\mathcal{C}_{\leftrightarrow}$:= {coop, sym, conf}. The whole set of representative labels is $\mathcal{C} := \mathcal{C}_{\rightarrow} \cup \mathcal{C}_{\leftrightarrow}$, with 15 unidirectional and 3 bidirectional labels, 18 in total. A label is a name describing a relation form; the definition fixes no establishment condition, threshold or normative value for any category. $\mathcal{C}_{\rightarrow}$ and $\mathcal{C}_{\leftrightarrow}$ are disjoint as label sets. The count 18 is the total of the 15 possible category-component values of a unidirectional descriptor and the 3 auxiliary bidirectional labels; it does not mean that all 18 become component values of one and the same unidirectional descriptor.

The manuscript gives a reading table of names and levels: observation (obs), reference (ref), affinity (aff), dependence (dep), support (sup), parasitism (par), domination (dom), management (mgmt), manipulation (manip), exploitation (exploit), use (use), competition (comp), alertness (alert), avoidance (avoid) and exclusion (excl) are unidirectional; cooperation (coop), symbiosis (sym) and conflict (conf) are bidirectional. Competition can be recorded in one direction, and mutual competition is treated as a same-kind mutual relation (`rem:competition_directionality`). In the table, "unidirectional" refers to the range of $\mathrm{cat}_{i \leftarrow j,t}$ and "bidirectional" to the range of $\mathrm{cat}_{\leftrightarrow,ij,t}$. The table is a correspondence for reading and not a criterion table for judging the establishment of an individual label.

Role of the category set (`rem:category_set_role`). $\mathcal{C}$ is a set of representative labels and not a classification system covering all relations. Giving the range of unidirectional labels does not mean that $\mathrm{cat}_{i \leftarrow j,t}$ is assigned to every object and time. The establishment condition of each label, inclusion relations among categories, transition rules and intensity scales are not fixed in this section. That one relation may carry several natural-language properties is distinguished from attaching one category label to a record at one time; formal rules for composite categories and multiple labels are left to later refinement.

No neutral category (`rem:neutral_exclusion`). This section does not add "neutral" or "contact" as a 19th representative label. Three cases are mutually distinct: the descriptor is not defined; the relation label cannot be judged; and the descriptor exists and all three differences are zero. From being unclassified, the non-existence, non-effect or neutrality of a relation is not inferred. undetermined is a value used only for the directionality indicator and is not a name for adding a relation category outside $\mathcal{C}$. From a record of "having made contact" alone, this paper does not conclude that the relation carries a specific cost or a change in feasibility-related quantities. Attaching no category is distinguished from constructing a six-component descriptor whose values are zero.

Definition (`def:six_system_classification`): six-family classification (classification C). For descriptive convenience the 18 labels of $\mathcal{C}$ are divided into six subsets:

Family	Symbol	Labels
recognition	$\mathcal{C}_{\text{recog}}$	obs, ref
affinity	$\mathcal{C}_{\text{aff}}$	aff
binding	$\mathcal{C}_{\text{bind}}$	dep, sup, coop, sym
control	$\mathcal{C}_{\text{ctrl}}$	dom, mgmt, manip
extraction	$\mathcal{C}_{\text{extract}}$	par, exploit, use
competition and isolation	$\mathcal{C}_{\text{comp-iso}}$	comp, alert, avoid, excl, conf

The six sets are disjoint as label sets and their union coincides with $\mathcal{C}$ (2+1+4+3+3+5=18). The partition follows from the enumeration of labels; it adds no establishment condition for categories and no ranking of values.

Role of the six families (`rem:six_system_role`). The families are an organization by content and not a ranking of superiority or of good and bad. A change in the direction of constraint can be recorded in the binding family, and a change in the direction of feasibility preservation in the competition and isolation family. Two records of the same category need not have the same three differences. Membership in a family gives neither a rule for computing the effect on each object nor a function giving the direction of change of feasibility. Dependence or support may appear as a record that widens the individual's choices under some conditions and as a record accompanying resource load or constraint under others. Likewise this paper does not state, from the names alertness or avoidance, a decrease of feasibility-related quantities as a general consequence. In each case the value of an individual difference and the causal contribution of a relation are separate problems.

Independence of the families and the establishment level (`rem:six_system_directionality`). The families are an organization by content and do not uniformly specify the level of establishment. The binding family contains unidirectional dependence and support and bidirectionally established cooperation and symbiosis; the competition and isolation family contains unidirectional competition, alertness, avoidance and exclusion and bidirectionally established conflict. The establishment level follows the distinction between $\mathcal{C}_{\rightharpoonup}$ and $\mathcal{C}_{\leftrightarrow}$ of `def:category_set`. Choosing one family does not determine whether a unidirectional or a bidirectional record should be made; one checks to which label set the category belongs.

Separation of categories from normative evaluation (`rem:category_neutral`). Cooperation, parasitism, domination, avoidance and similar names are labels describing structural relations; the name alone fixes neither desirability, harmfulness nor the direction of change of feasibility. This paper does not state, from a category name or family membership, the truth or falsity of the time-indexed satisfaction predicates of SM, EU and FR of §3 (`sec:three_principles`) as a general consequence. For the interpretation of a relation versus the internal recognition of the ACS itself, the distinction of `rem:recognized_relation_category` is used. Not giving a normative evaluation does not mean that it is immaterial whether the differences attached to a relation record are positive or negative; recording the sign of a difference is separated from assigning ethical good or bad to a category name.

Undefined status of category conditions (`rem:category_phase1a`). Enumerating the 18 representative labels differs from giving the establishment condition of each. This section does not define characterizing conditions, judgment thresholds, durations, intensity scales, temporal subtypes or transition rules among categories; their deterministic refinement is left to Phase 1.α. For example, "how much resource load makes a relation exploitation" and "how much mutuality makes it cooperation" are not judgment rules derived from the label set. Temporal subtypes such as long-term, cumulative support are likewise not determined by the one-time labels of this section.

Derived and composite categories (`rem:derived_composite_categories`). Entrustment, exchange, membership, friendship and similar notions may be describable as combinations of several relation categories, but this section does not fix a rule of construction. Temporal change from one category to another is not generated from the fixed descriptive form of this section alone. Definitions and transition rules of derived and composite categories are left to

Phase 1.α. For instance, relations such as entrustment or exchange may possibly be described by combining records of reference, dependence, support, management or use, but this paper does not make such combinations formal category definitions. Changes of classification such as from support to dependence, or from alertness to conflict, are not deduced from a one-time relation descriptor alone.

Categories and recognized relations (`rem:recognized_relation_category`). $\mathrm{cat}_{i\leftarrow j,t}$ is a descriptive label in the theory, and it is not assumed that Σ_i recognizes that relation under the same label. This distinction is consistent with the role difference between internal identification in §4 (`sec:tipa`) and the descriptive classification of this section. Indicators of a recognized relation such as $\mathrm{recog}_{i\leftarrow j,t}$ are listed here only as symbol candidates; their range, computation rule and correspondence with categories are not defined. Hence from $\mathrm{cat}_{i\leftarrow j,t}$ this paper does not infer the content or correctness of the ACS's own recognition, and the refinement is left to Phase 1.α. How a relation that the individual recognizes as support is to be recorded in theory is not determined unless the correspondence between internal recognition and external description is defined separately. $\mathrm{recog}_{i\leftarrow j,t}$ in this paper is a name that announces that refinement.

Categories and the section on blockage and collapse (`rem:categories_to_sec7`). Membership in the 18 categories or the six families is not a condition of belonging to $\mathfrak{P}_{\mathrm{blk}}$ or $\mathfrak{P}_{\mathrm{col}}$ of §7 (`sec:collapse_cond`). Even if feasibility-related quantities change along with a relation record, the judgment of blockage or collapse requires separate conditions on state quantities and trajectory records. The local size indicator being zero is not identified with $A_{i,t} = \emptyset$. Even for the binding or affinity family, the possibility that a difference in the direction of constraint is recorded depending on the individual's conditions is not excluded. Conversely, from a record with a label of the competition and isolation family, reachability of blockage or collapse is not directly derived. The connection with path sets depends on the time domains and trajectory conditions specified in later sections.

Out-of-scope matters, hand-off to later sections, and summary

The subsection distinguishes the established descriptive skeleton from what is delegated to each Phase, and organizes the hand-off to §7 and §8.

Minimal structure established in this section (`rem:sec6_established`). What §6 gives is: the descriptive three-way classification of external objects; the six-component unidirectional relation descriptor, which is well-formed under conditions; the bidirectional composition between ACSs with an auxiliary directionality; the relation quality relative to an evaluation rule; and the 18 representative categories with the six-family classification. The local indicators, finite-window differences and six-component descriptors are forms for constructing a description when the domain of application and values are given; they are not an existence theorem that every relation can be quantified. This section does not establish that descriptor values exist at each time, the establishment conditions of categories, or the specific causal effect of a relation.

Items delegated to Phase 1.α (`rem:phase1a_delegation`). Within the deterministic, discrete-time framework, the following refinements are delegated (`subsec:phase1a`). The names listed are subjects for examination and are not present definitions or established theorems.

1. Establishment conditions and thresholds of each of the 18 categories (`def:category_set`, `rem:category_phase1a`).
2. Intensity and duration of a relation, and temporal subtypes such as long-term, cumulative support.
3. Cancellability, substitutability, mediation and termination conditions of a relation.
4. Response-side categories such as being dominated, managed, manipulated, exploited or used (`rem:homo_hetero_rel`).
5. Composite description rules when domination, parasitism and similar appear bidirectionally (`rem:no_mutual_form`).
6. Derived and composite categories such as entrustment, exchange, membership and friendship, and transitions among categories (`rem:derived_composite_categories`).
7. Correspondence between relations recognized inside the ACS and categories in theory

(rem:recognized_relation_category).

8. Extended forms describing bidirectional interaction with non-ACS agents and similar objects (rem:nACS_limit).
9. A quantitative criterion for "main" in the directionality indicator $\mathrm{dir}_{ij,t}$ (rem:dir_undefined).
10. The evaluation domain and concrete form of the local size indicator sz_i, concretization by finite approximation or covering numbers at fixed precision, and, where needed, additional conditions for being size-like (def:rel_size_functional).
11. Bidirectional establishment rules for cooperation, symbiosis and conflict, their correspondence with unidirectional categories, and a quantitative criterion distinguishing the unidirectional and mutual forms of competition (def:bidirectional_category, rem:competition_directionality).

The concrete procedure for object identification and the rules that generate and update relation records over finite windows are also treated on the refinement side where needed. These refinements are not used as if already defined in this section.

Items delegated to Phase 2 (rem:phase2_delegation). Phase 2 receives probabilistic establishment, transition and termination; continuous time; interaction involving the internal dynamics of external objects; and social network dynamics (subsec:phase_outlook). The description of objects whose classification map changes with time is also beyond the fixed classification of this section. Interaction involving the internal state and internal dynamics of an external object, and processes in which several relations change over a network, are not specified by the unidirectional and bidirectional descriptors of this section alone. Attaching time indices in this paper is not to be read as introducing probabilistic transition laws or network dynamics, and the delegation does not mean regarding the differences in individual record windows as probabilistic quantities.

Hand-off to the sections on blockage, collapse and the organizational ACS (rem:sec6_to_sec78). §7 (sec:collapse_cond) treats blockage, collapse and the respective reaching paths, and §8 (sec:org_acs) treats the conditions under which an ACS structure holds for the organization itself, the organizational ACS. The relation descriptors of this section are reference material for both sections; they are neither a judgment of a failure state nor a sufficient condition for an organizational ACS to hold. For the connection to §7, the three differences attached to a relation record are distinguished from the blockage and collapse conditions on an individual's trajectory. For the connection to §8, the recording of an ordered pair of relations is distinguished from the holding of an organization-level ACS structure for the organization itself. This paper does not state, from this section read alone without those later sections, the judgment of either as a general consequence.

Summary of the descriptive policy of this section (rem:sec6_summary). The relation categories, observation-window differences and local size indicators of this section are value-neutral structural descriptions. They do not prove normative judgment, a contribution that is causal for each object, the reaching of blockage or collapse, or the holding of an organizational ACS. This paper also does not state, as a general consequence, that the magnitude of an indicator value can be read as an inclusion relation between feasible-action sets, or that relation descriptors are defined for all external objects at all times. When differences are compared, the window length and the scale of the indicator are fixed; in bidirectional records, the record-valid time domain and the relation-continuation condition of each individual are distinguished. Connection with later sections is made only under the domains and conditions that each section separately specifies.

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.